

Infrastructure capacity assessment

What this tool produces · *A spatial diagnostic report highlighting specific urban nodes where the current capacity of utilities and mobility infrastructure falls short of the actual or projected land occupation.*

G1 · Proof of Impact

Dimensions

Spatial

Economic

Environmental

Preparing the workshop · *Infrastructure capacity assessment*

G1 · Proof of Impact

Why this tool matters

Looking beneath the visible surface of the city to respect its vital organs; ensuring that the ambition of growth does not fatally rupture the hidden lifelines that sustain everyday survival.

Evidence we produce

A spatial diagnostic report highlighting specific urban nodes where the current capacity of utilities and mobility infrastructure falls short of the actual or projected land occupation.

Duration & Material

Duration · 2 to 4 months of cross-departmental data analysis

Material · GIS software, municipal utility blueprints, demographic projections, digital twin platforms

Reference case

The UN-Habitat Incremental and Participatory Toolbox for Urban Planning utilizes capacity mapping to identify urban nodes where physical infrastructure capacity clashes with actual land occupation. By explicitly assessing structural vulnerabilities, transit capacities, and water networks, planners can target precise, incremental interventions to upgrade the city's lifelines before approving densification.

Suggested agenda · Brief (10 min) > Method walk-through (15 min) > Animation canvas (60-90 min) > Harvest (20 min)

Networks × current load × future demand × headroom

Duration
2 to 4 months of cross-departmental data analysis

Water & sewer supply, treatment, storm	Energy electricity, gas, district	Roads & transit capacity, modal split
Telecoms fibre, mobile, 5G	Solid waste collection, treatment, recycling	Public services schools, health, civic

For each network · current load · future demand · headroom remaining.

Harvesting our *learnings*

After running the canvas, capture three layers of insight.

What inspires me

Stand-out signals, surprising stories, ideas to remember

How to apply it here

Concrete adaptations to our context, our team, our constraints

What we need to be autonomous

Skills, allies, resources, decisions still to secure

Each participant fills a copy. Then debrief together to surface patterns and next steps.